High-Performance In-Memory Data Wrangling Using data.table

Project Title

High-Performance In-Memory Data Wrangling on Enterprise Sales Transactions using data.table

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Project Objective

The objective of this project is to understand how enterprise organizations process and analyze extremely large transactional datasets using the **data.table** package in R. The project demonstrates high-speed data manipulation using pointer-level reference semantics, enabling efficient in-memory updates, filtering, sorting, indexing, grouping, and aggregation. The project also includes business analysis, report generation, and visualization while handling a dataset containing **500,000 transaction records**.

Software Used

- R Programming
- RStudio
- data.table Package
- Git
- GitHub

Dataset Description

A synthetic enterprise sales transaction dataset containing **500,000 records** was generated using R.

The dataset includes the following columns:

- TransactionID
- CustomerID (Renamed as CustID)
- CustomerName
- Gender
- Age
- ProductID (Renamed as ProdID)
- ProductName
- Category
- Quantity
- UnitPrice
- Discount

- GST
- PaymentMethod
- City
- State
- OrderDate
- SalesPerson
- Revenue
- DiscountAmount
- GSTAmount
- NetRevenue

The dataset was saved as:

sales_transactions.csv

Project Tasks

Task 1 – Generate Dataset

- Generated 500,000 transaction records
- Saved dataset as **sales_transactions.csv**
- Displayed first and last 10 records
- Displayed dataset dimensions
- Displayed dataset structure

Task 2 – Load and Explore Dataset

- Imported dataset using fread()
- Displayed first and last records
- Displayed column names
- Checked data types
- Displayed summary statistics
- Identified missing values
- Checked duplicate records
- Counted unique customers, products, and cities

Task 3 – Data Cleaning and Preparation

- Removed duplicate records

- Renamed columns
- Reordered columns
- Converted data types
- Displayed cleaned dataset

Task 4 – High-Speed Data Manipulation

- Selected single and multiple columns
- Filtered rows using conditions
- Sorted data
- Displayed Top 20 expensive products
- Displayed Top 20 highest quantity transactions

Task 5 – In-place Column Updates

Created new columns using the := operator:

- Revenue
- DiscountAmount
- GSTAmount
- NetRevenue

Task 6 – Index Keying

- Created keys using setkey()
- Indexed Customer ID
- Indexed Product ID
- Performed indexed searches
- Compared search performance

Task 7 – Grouping and Aggregation

Generated:

- Customer Summary
- Product Summary
- City Summary
- Category Summary

Task 8 – Business Analysis

Solved business questions related to:

- Highest revenue city

- Maximum sales product
- Highest ordering customer
- Most used payment method
- Average transaction value
- Highest revenue category
- Top 20 customers
- Top 20 products

Task 9 – Export Reports

Exported:

- customer_summary.csv
- product_summary.csv
- city_summary.csv
- category_summary.csv

Task 10 – Visualization

Generated Base R visualizations:

- Transactions by City
- Revenue by Product Category
- Top 20 Customers
- Payment Method Distribution

Steps to Run the Program

1. Install R and RStudio.
2. Install the data.table package.
3. Open the file **high_performance_data_wrangling.R**.
4. Run the script from beginning to end.
5. The program will:
    - Generate the dataset
    - Save the dataset as sales_transactions.csv
    - Perform data exploration and cleaning
    - Generate business reports
    - Export CSV files

- Create visualizations
- Save outputs in the **Output** folder.

GitHub Repository Structure

High-Performance-DataWrangling-Week2/ | ├── README.md ├── Week2_Report.pdf ├── sales_transactions.csv ├── high_performance_data_wrangling.R | └── Output/ ├── city_transactions.png ├── category_revenue.png ├── top_customers.png ├── payment_distribution.png

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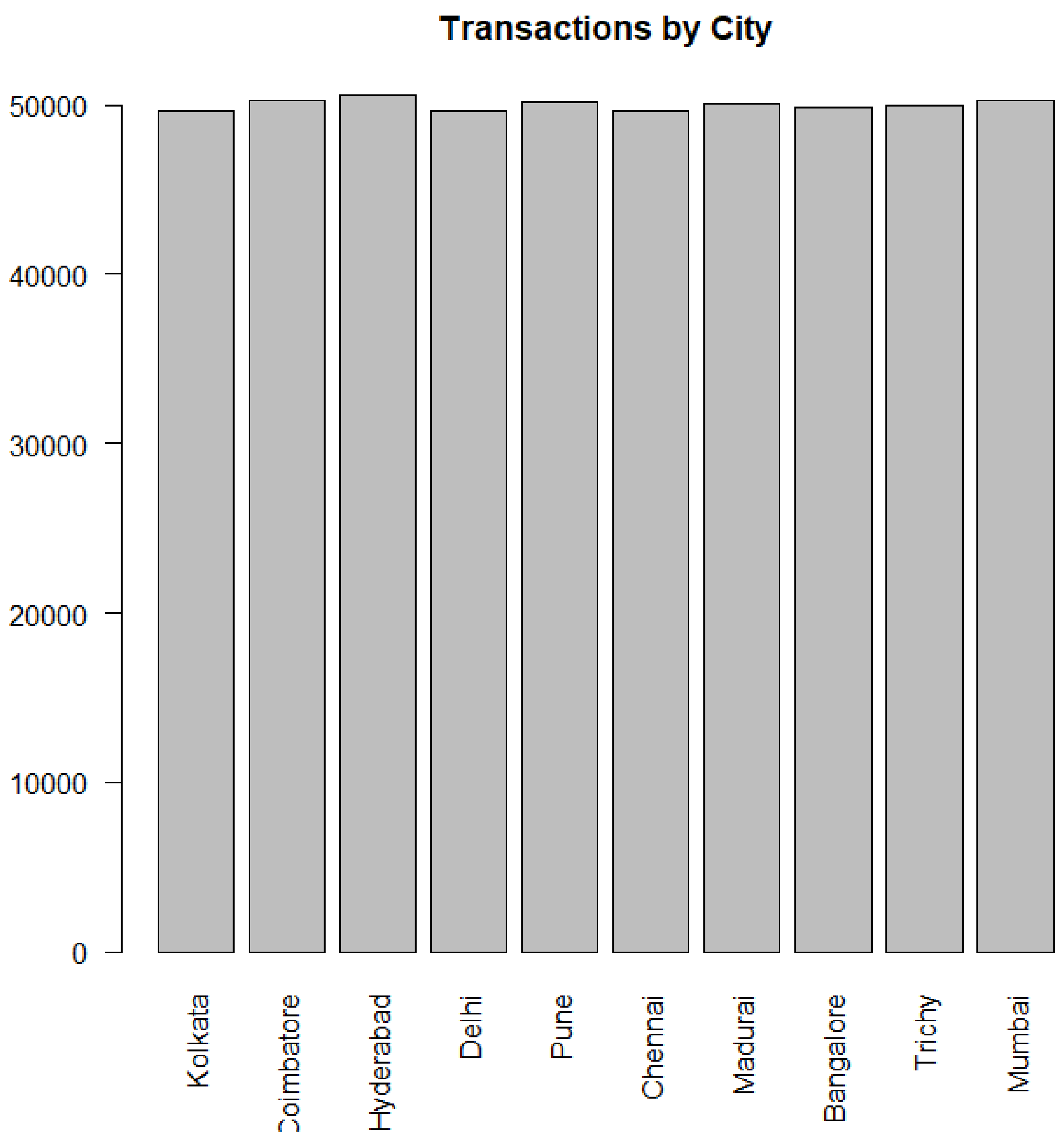
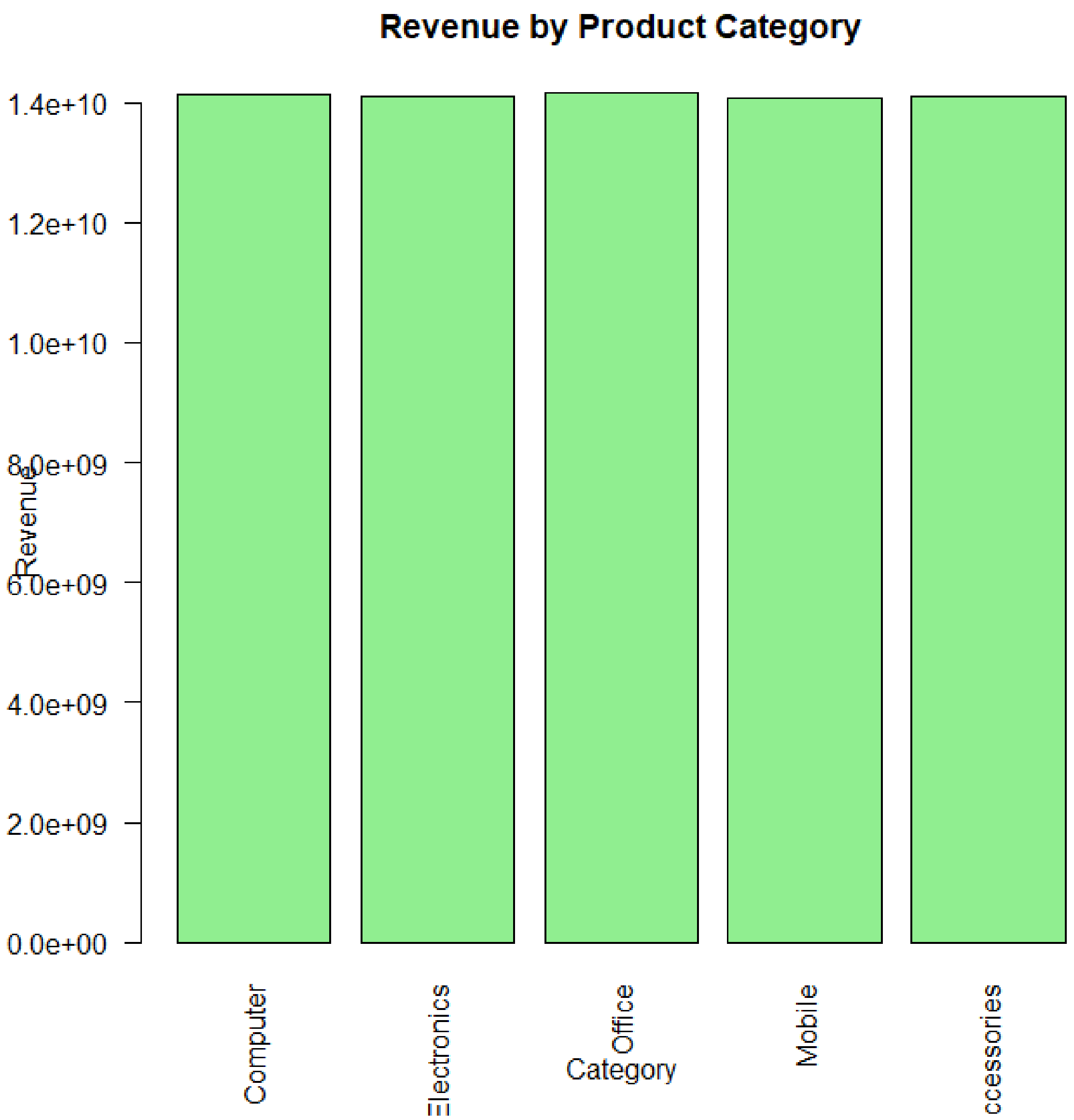
Output Screenshots

Transaction by City:



Revenue by Porduct Category:



Top 20 Customers by Revenue:

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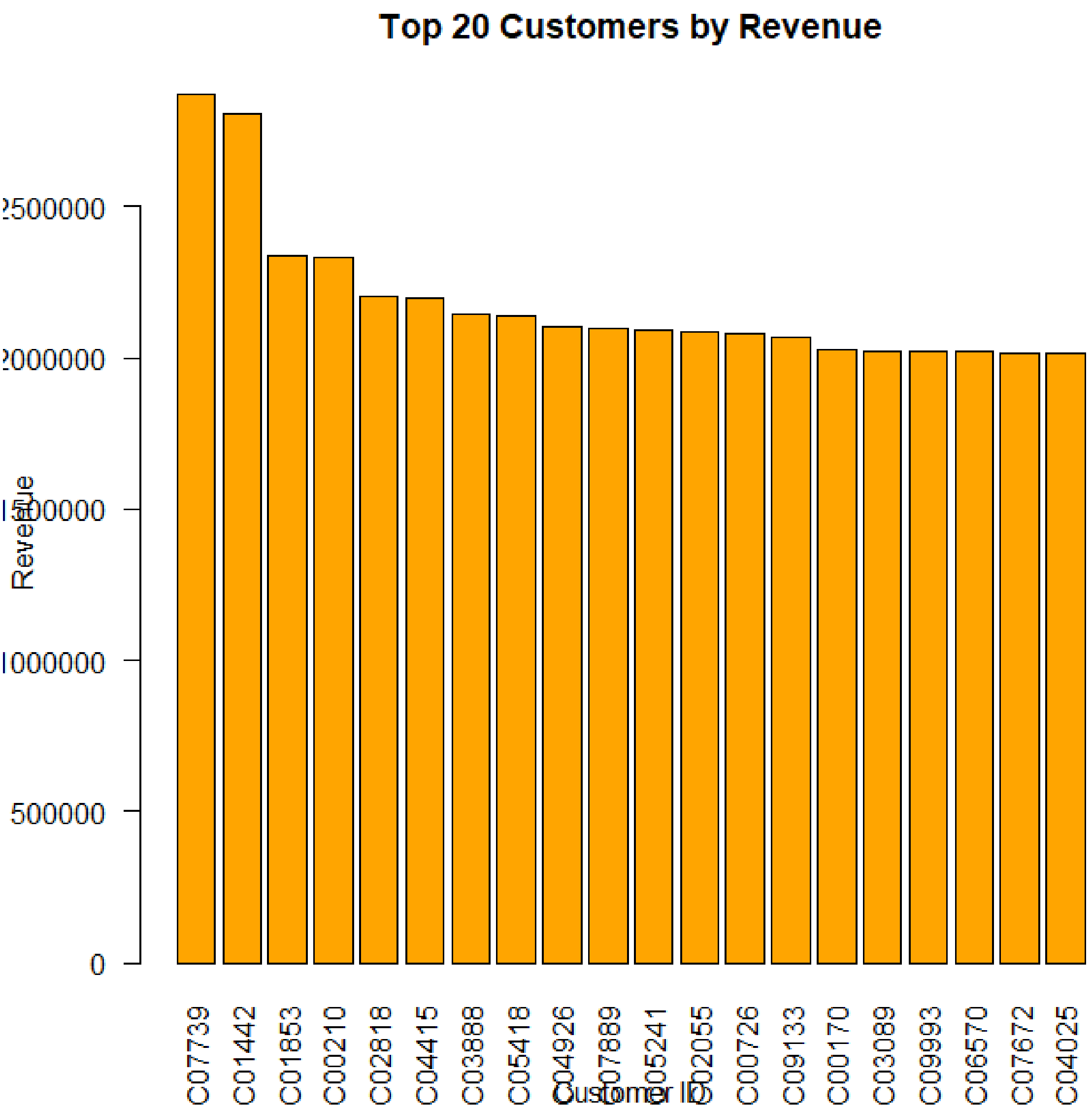
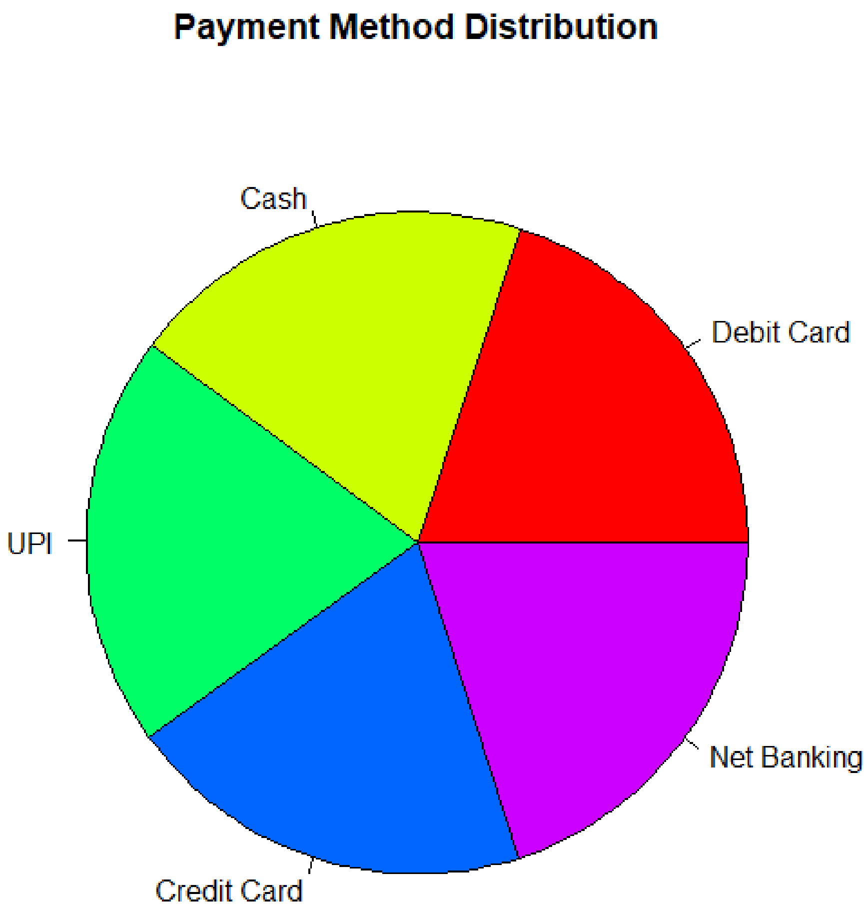


Payment Method

Distribution:



Business Questions Solved

- Which city generated the highest revenue?
- Which product generated the maximum sales?
- Which customer placed the highest number of orders?
- Which payment method is used most frequently?
- What is the average transaction value?
- Which category generated the maximum revenue?
- Who are the Top 20 customers by revenue?
- Which are the Top 20 products by quantity sold?

Learning Outcomes

After completing this project, I learned to:

- Generate large synthetic datasets in R.
- Use the **data.table** package for efficient data manipulation.
- Perform in-place updates using the := operator.
- Apply indexing with setkey() for faster searching.
- Perform grouping and aggregation operations.
- Analyze enterprise transaction data.
- Export reports in CSV format.
- Create visualizations using Base R.
- Organize project files and manage version control using Git and GitHub.

This version follows the exact sections requested in your project description and is suitable to save as **README.md**.